

WEEK 1 DAY 4

NAME:VUDA JAYA SAI GANESH

GROUP:S2

Dataset Design & Quality Expectations

1. Expected Dataset Schema & Structure

1.1 File Format & Organization

- **Dataset Name:** SOC DATA SET
- **Source:** Faculty-provided (TIH IIT Guwahati)
- **Format:** ZIP archive containing CSV files
- **Structure:**
 - Single CSV file or multiple CSV files (organized by session)
 - Each row represents a single timestamp measurement during a charging session

1.2 Expected Columns & Definitions

Column Name	Data Type	Description	Valid Range	Format	Notes
session_id	String/Int	Unique identifier for each charging session	Unique	e.g., S001	Critical for splitting
timestamp	Datetime	Time of measurement	Monotonic	YYYY-MM-DD HH:MM:SS	Frequency unknown
voltage	Float	Battery voltage	3.0–4.5 V	e.g., 3.72	Key feature
current	Float	Charging current	0–100 A	e.g., 15.3	Shows charging profile
power	Float	Charging power	0–10 kW	e.g., 2.8	Derived from $V \times I$
temperature	Float	Battery temperature	-10–60°C	e.g., 25.5	Safety-critical
soc	Float	State of Charge (Target)	0–100%	e.g., 45.2	Primary prediction target

Column Name	Data Type	Description	Valid Range	Format	Notes
charge_time	Float	Remaining Charging Time (Target)	0–∞ minutes	e.g., 35.0	Secondary target

1.3 Potential Additional Columns

- **cycle_number** – Battery cycle count (Medium priority)
- **ambient_temp** – Ambient temperature (Medium priority)
- **battery_id** – Battery identifier (Low priority)
- **charging_profile** – Fast/slow charging type (Low priority)
- **health/SOH** – State of Health indicator (High priority)

1.4 Metadata Documentation Template

A structured template is provided to record dataset schema, column definitions, and data quality notes for reproducibility and clarity.

2. Dataset Risk Checklist

2.1 Missing Data Risks

- **M1:** Complete columns missing → Mitigation: contact mentor
- **M2:** Gaps in time series → Mitigation: interpolate/drop
- **M3:** Entire sessions missing → Mitigation: drop sessions
- **M4:** Incomplete sessions (<100% SoC) → Mitigation: use carefully
- **M5:** Missing timestamps → Mitigation: drop rows

2.2 Data Quality Risks

- Outliers, noise, invalid values, duplicates, inconsistent units, and type mismatches are anticipated.
- Mitigation strategies include smoothing, range checks, and standardization.

2.3 Data Leakage Risks

- **Critical:** Ensure splitting by `session_id` to avoid leakage.
- Avoid using future information, target-related features, or scaling before splitting.

2.4 Imbalance Risks

- Sparse data at low/high SoC ranges.
- Weighted loss functions and stratified sampling recommended.

2.5 Temporal Risks

- Non-uniform sampling, shuffled rows, and varying session durations.
- Resampling and normalization required.

2.6 Annotation/Labeling Risks

- Incorrect SoC calculation or inconsistent definitions.
- Must validate with domain knowledge and mentor guidance.

3. Train/Validation/Test Split Strategy

Correct Approach

- Split by **`session_id`** (70% train, 15% validation, 15% test).
- Ensure no session appears in more than one split.
- Preserve temporal order within sessions.

Implementation (Pseudo-code)

python

```
sessions = df['session_id'].unique()
```

```
np.random.seed(42); np.random.shuffle(sessions)
```

```
train_sess = sessions[:0.7*n]
```

```
val_sess = sessions[0.7*n:0.85*n]
```

```
test_sess = sessions[0.85*n:]
```

```
train = df[df['session_id'].isin(train_sess)]
```

```
val = df[df['session_id'].isin(val_sess)]
```

```
test = df[df['session_id'].isin(test_sess)]
```

4. Pre-Split Data Quality Check Plan

- **Before Splitting:** Check missing values, duplicates, outliers, invalid ranges, and session counts.
- **After Splitting:** Verify distribution similarity, SoC coverage, session independence, and target balance.

5. Data Handling Strategy

- **Missing Values:** Interpolate small gaps, drop large gaps or incomplete sessions.
- **Outliers:** Detect via IQR/Z-score; cap or remove.
- **Scaling:** Use StandardScaler/MinMaxScaler fitted only on training data.

6. Pre-Split vs Post-Split Analysis

- Pre-split: overall quality, missing values, correlations, leakage risks.
- Post-split: verify distributions, leakage prevention, and consistency across splits.

7. Evidence & Submission

- **Deliverables:** Dataset schema, risk checklist, split strategy documentation.
- **Evidence Links:**
 - Google Drive Folder:
https://drive.google.com/file/d/1ZqbmD1q1BDimEayC3wIBicL0ijhBUULX/view?usp=drive_link

8. Summary

- Dataset expected in CSV format with key features: voltage, current, power, temperature, SoC, and charge time.
- Risks include missing data, outliers, leakage, imbalance, and annotation errors.
- Correct split strategy is by session_id to ensure independence.
- Pre- and post-split checks are essential for robust model training.